MD-GBM · model specification

V4 1.5-year monthly 10,000-path study · documentation source of truth is the companion notebook

Name

- MD-GBM means Markov Directional Geometric Brownian Motion.

Idea

- MD-GBM is original Modified GBM with one calibration fix. It is the return-based model in the V4 monthly 10,000-path study. Price map $S_{t+1}=S_t e^{R_t}$. Direction under P is the 1-lag up/down Markov chain. Sizes are $|N(\mu, \sigma^2)|$ on up and down bars. Q is the usual additive shift so $E[e^R]=e^{r_f \Delta t}$.
- Only change: μ and σ are inverted so $E[|N(\mu, \sigma)|]$ equals the sample mean of $|R|$ in that bucket. Original set $\mu = \text{mean}(|R|)$ and then drew $|N(\mu, \sigma)|$, so simulated sizes were too large. If the half-normal at the sample SD already overshoots, $\mu=0$ and σ is shrunk to match the mean.

What it is not

- Not Merton and not GARCH.

MD-GBM · estimation, Q, and files

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Estimation (each 18-month lookback)

- Drop zero log-returns. Laplace-smoothed P(U|U) and P(D|D) from consecutive sign pairs.
- On up (down) bars, invert folded-normal μ, σ so $E[|N|]$ equals the sample mean of $|R|$.
- Simulator start: last observed non-zero sign. Parameters dated t are used only after t.

Simulation

- P-measure (stock PDF): no drift shift. Path cloud n=10,000, seed 42. Reported path = p50 vs realized adj-close.
- Q-measure (LSM): after drawing R, add $r_f \Delta t - \log E[e^R]$. Then Longstaff–Schwartz on the same Monday ATM listed-call sample as every other model.

Parameters in the estimation PDF

- P(U|U), P(D|D), P(U|D), P(D|U), μ_U, σ_U, μ_D, σ_D.

Where results are

- LSM ranking: results/1p5y_monthly_return_based/01_optimal_stopping/.
- Code: code/md_gbm/20*_modified_gbm_meanfix.ipynb.